GATE

ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 07



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Recap of Previous Lecture



Topic Profit Loss; Mixtures Alligations

Topics to be Covered



Topic-1 More on Mixtures Alligations

Topic-2 Number System (Basics)

[MCQ]

#Q. If the side of a square is increased by 20%, then what is the percentage change in its area?

Assignment

A 44%

B 80%

C 22%

D 144%

Area = $S \times S$

$= 1.2 \times 1.2 = 1.44$

.44

44%

[MCQ]

Twice

100%↑

#Q. The population of a town doubled every 5 years from 2000 to 2015. What is the percentage increase in population in this period?

Assignment

A 800%

B 400%

C 700%

D 600%

$2 \times 2 \times 2 = 8$

1 - 2000 - 2005

1 - 2005 - 2010

1 - 2010 - 2015

7 ∩

700%↑

[MCQ]

#Q. If P is 60% taller than Q, by what percent is Q shorter than P?

Assignment

$P = 160\% \text{ of } Q$

$\frac{100}{160} P = Q$

$\Rightarrow 0.625P = Q$

0.375

37.5%

A 40%

B 37.5%

C 62.5%

D None of these

[MCQ]

#Q. A is twice B and B is 200% more than C. By what percent is A more than C?

Assignment

$A = 2B$

$B = 3C$

$A = 2 \times 3C$

$A = 6C$

500%

A 200%

B 400%

C 500%

D 600%

[MCQ]

#Q. The population of a village is 5500. If the number of males increases by 11% and the number of females increases by 20%, then the population becomes 6330. The population of the female in the village is?

A 2000

B 2500

C 3000

D 3500

$M + F = 5500$

$1.11M + 1.2F = 6330$

$1.11M + 1.11F = 6105$

$0.09F = 225$

$\therefore F = \frac{225}{0.09} = \frac{22500}{9}$

Assignment

[MCQ]

#Q. 5% of income of P is equal to 15% of income of Q and 10% of income of Q equal 20% of income of R. If R's income is 2000, then What is total income of P, Q and R?

A 9000

B 12000

C 15000

D 18000

$5\% P = 15\% Q$

$P = 3Q$

$10\% Q = 20\% R$

$Q = 2R$

Assignment

$P = 12000$

$Q = 4000$

$R = 2000$

$18,000$

PROFIT & LOSS



[MCQ]

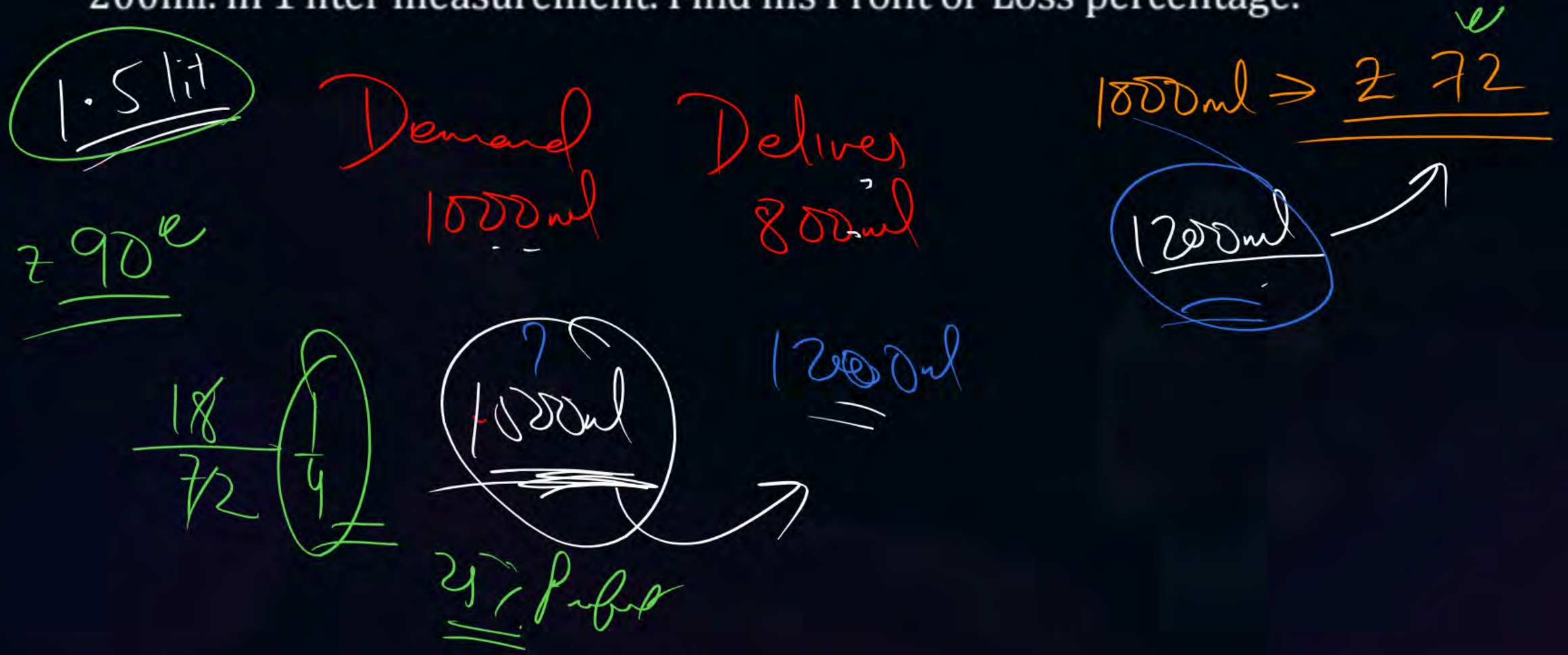
#Q. A milk vendor purchases milk at Rs. 72/ litre, and sells at Rs. 60/ litre. For every 1 litre milk he adds 200ml. of water. While selling milk he cheats 200ml. in 1 liter measurement. Find his Profit or Loss percentage.

$$\frac{S.P.}{C.P.} = \frac{60}{72} \times \frac{1200}{1000} \times \frac{1000}{800} = \frac{5}{4} = 1.25$$

25% Profit

[MCQ]

#Q. A milk vendor purchases milk at Rs. 72/ litre, and sells at Rs. 60/ litre. For every 1 litre milk he adds 200ml. of water. While selling milk he cheats 200ml. in 1 liter measurement. Find his Profit or Loss percentage.



[MCQ]

#Q. A cloth merchant purchases cloth at ₹80/meter and sells at ₹100/meter. As a festive offer, he gives 50% extra free on every demand. Find his profit% or loss%.

A. 25% profit

B. 8.33% profit

C. 16.66% loss

D. 83.33% profit

[MCQ]

#Q. Due to downfall in the market, 'A' sells mangoes to 'B' saying "FOR EVERY 12 MANGOES, COUNT AS 8". Due to overnight demand, 'A' took his mangoes back from 'B' saying " FOR EVERY 8 MANGOES, COUNT AS 12". Find the profit or loss percent of 'A'.

$55.5\% \uparrow_{108}$

$\frac{8}{12} \times \frac{12}{8} = 1 = \frac{4}{9} = 0.\overline{4}$

0.5 ×

PROFIT & LOSS

Note:

A trader may sometimes have multiple profits or losses simultaneously.

This is equivalent to having multiple changes and so all individual changes are to be multiplied to get the overall effect.

[MCQ]

#Q. An alloy A consist of iron & copper in the ratio 5 : 7, whereas alloy B consist iron & copper in the ratio 3 : 1. In what ratio these two alloys are to be mixed to form a new alloy with iron & copper in the ratio 2 : 1?

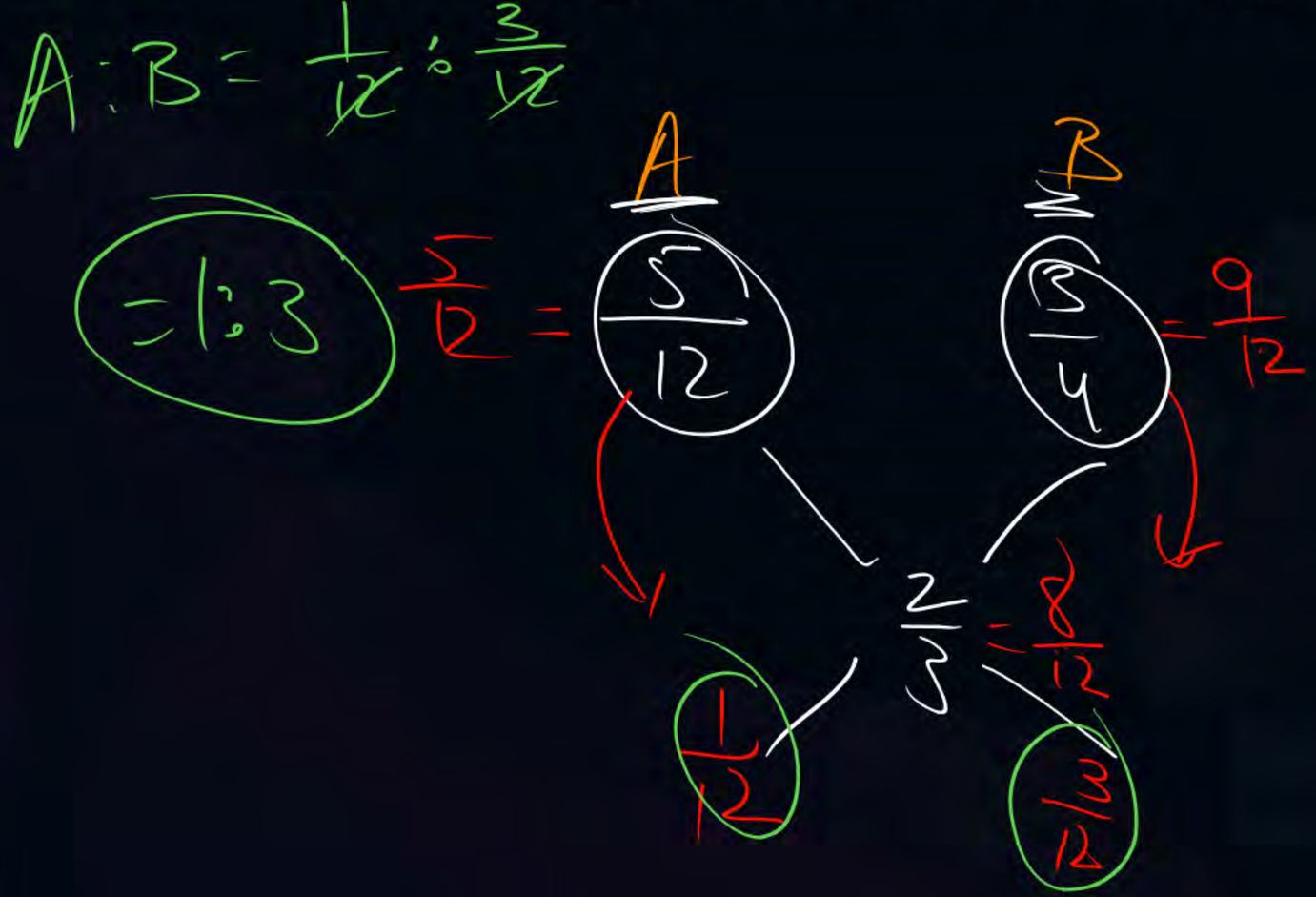
Assignment

A 5:13

B 7:5

C 1:3

D 3:1



[MCQ]

#Q. How many litres of water must be added to 16 litres milk containing 10% water to dilute it with 20% water?

A. 3 litres

B. 2 litres

C. 1 litre

D. 4 litres

16 lit

Water 1.6 lit Milk 14.4 lit

$$\frac{1.6+x}{16+x} = \frac{20}{100}$$

Assignment



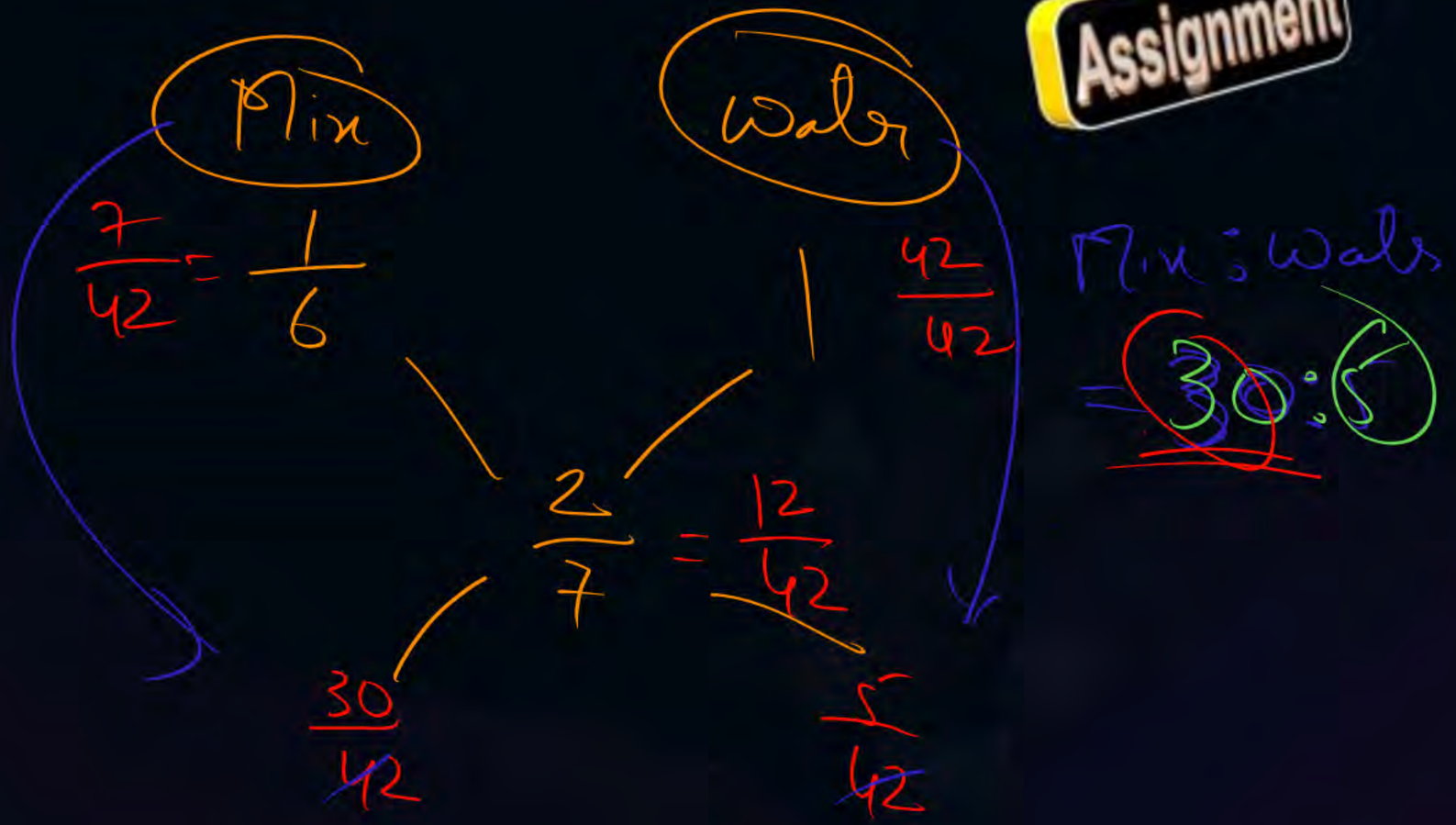
Mix : water = 80 : 10 = 8 : 1

= 16 : 2

[MCQ]

#Q. A mixture contains milk and water in the ratio 5:1. On adding 5 liters of water, the ratio of milk to water becomes 5:2. the quantity of milk in the mixture earlier was:

Assignment



A 16 litres

B 25 litres

C 32.5 litres

D 22.75 litres

[MCQ]

#Q. In what ratio must a grocer mix two varieties of pulses costing ₹15 and ₹20 per kg respectively so as to get a mixture worth ₹16.50 per kg?

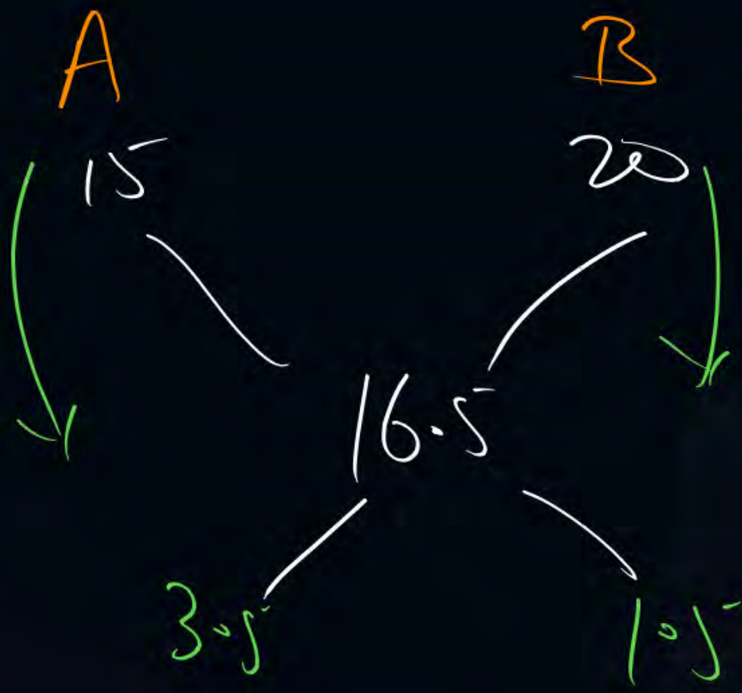
Assignment

A 3:8

B 5:3

C 7:3

D 4:5



7 3

A : B = 3.5 : 1.5

= 7 : 3



2 mins Summary

Topic ~~Number System~~

Profit loss & Mixtures

Practice Questions

THANK - YOU